

Homework 10

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Problem 1a.

Proof. Let $f(x) := x^2 + 2x$. $\forall \epsilon > 0$. Let $\delta = \min(1, \frac{\epsilon}{7})$. $\forall x \in \mathbb{R} \ni |x - 2| < \delta$. By the reverse triangle inequality, $||x| - 2| \leq |x - 2| < \delta \implies -\delta < |x| - 2 < \delta \implies |x| < \delta + 2 \leq 1 + 2 = 3$. Thus $|x| < 3$. Thus,

$$\begin{aligned} |f(x) - 8| &= |x^2 + 2x - 8| = |x + 4| |x - 2| \\ &< \delta |x + 4| \quad (\text{since } |x - 2| < \delta) \\ &< \delta(|x| + 4) \quad (\text{triangle inequality}) \\ &< 7\delta \quad (\text{since } |x| < 3) \\ &\leq \epsilon \quad (\text{since } \delta \leq \frac{\epsilon}{7}) \end{aligned}$$

Thus, $\lim_{x \rightarrow 2} f(x) = 8$

□

Problem 1b.

Proof. Let $f(x) := \frac{1}{1-x}$. $\forall \epsilon > 0$. Let $\delta = \min(\frac{1}{2}, \frac{\epsilon}{2})$. $\forall x \in \mathbb{R} \ni x \neq 1 \ni |x - 2| < \delta$.

Thus, $|x - 2| < \frac{1}{2} \implies |x - 1| > \frac{1}{2} \implies \left| \frac{1}{x-1} \right| < 2$.

$$\begin{aligned} |f(x) + 1| &= \left| \frac{1}{1-x} + 1 \right| = \frac{|x-2|}{|x-1|} \\ &< 2\delta \quad (\text{since } |x-2| < \delta \text{ and } \left| \frac{1}{x-1} \right| < 2) \\ &\leq \epsilon \quad (\text{since } \delta \leq \frac{\epsilon}{2}) \end{aligned}$$

Thus, $\lim_{x \rightarrow 2} f(x) = -1$

□

Problem 1c.

Proof.

$$\begin{aligned} |x| &:= \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases} \\ \sqrt{x^2} &:= \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases} \end{aligned}$$

Thus $|x| = \sqrt{x^2}$.

Thus $f(x) = \frac{x^2}{|x|} = \frac{x^2}{\sqrt{x^2}} = \sqrt{x^2} = |x|$.

$\forall \epsilon > 0$. Let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni x \neq 0 \ni |x| < \delta$.

Thus,

$$|f(x) - 0| = \left| \frac{x^2}{|x|} \right| = ||x|| = |x| < \delta = \epsilon$$

Thus, $\lim_{x \rightarrow 0} f(x) = 0$

□

Problem 1d.

Proof. Let $f(x) := \frac{x+3}{2x+5}$. $\forall \epsilon < 0$. Let $\delta = \min(1, 3\epsilon)$. $\forall x \in \mathbb{R} \ni |x+1| < \delta$. Thus, $|x+1| < 1$. $\forall a, b \in [-2, 0] \ni a < b$ thus $6a+15 < 6b+15$. Moreover $6(-2)+15 = 3 > 0$. Thus, $\forall x \in \mathbb{R} \ni |x+1| < \delta \leq 1$, since $x \in [-2, 0]$, thus $0 < 6(-2)+15 = |6(-2)+16| = 3 < 6x+15$, so $3 < |6x+15|$. So $\forall x \in \mathbb{R} \ni |x+1| < 1$:

$$\begin{aligned} \left| \frac{x+3}{2x+5} - \frac{2}{3} \right| &= \left| \frac{3x+9-4x-10}{6x+15} \right| \\ &= \frac{|-x-1|}{|6x+15|} \\ &= \frac{|x+1|}{|6x+15|} \\ &\leq \frac{|x+1|}{3} \\ &< \frac{\delta}{3} \\ &\leq \epsilon \end{aligned}$$

Thus, $\lim_{x \rightarrow -1} f(x) = \frac{2}{3}$

□

Problem 2a.

Proof. For all $\epsilon > 0$. Let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni |x-2| < \delta$

$$|x-2| < \delta = \epsilon$$

Thus $\lim_{x \rightarrow 2} x = 2$.

$\lim_{x \rightarrow 2} x^2 = \lim_{x \rightarrow 2} (x * x)$. By product of limits, $\lim_{x \rightarrow 2} (x * x) = (\lim_{x \rightarrow 2} (x)) (\lim_{x \rightarrow 2} (x)) = 2 * 2 =$

4. Thus $\lim_{x \rightarrow 2} x^2 = 4$.

By the theorem about the scalar multiplication of limits, $\lim_{x \rightarrow 2} 2x = 2 \lim_{x \rightarrow 2} x = 2 * 2 = 4$.

By the theorem about sum of limits, $\lim_{x \rightarrow 2} (x^2 + 2x) = \lim_{x \rightarrow 2} x^2 + \lim_{x \rightarrow 2} 2x = 4 + 4 = 8$ □

Problem 2b.

Proof. $\forall \epsilon > 0$, Let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni |x - 2| < \delta$

$$|x - 2| < \delta = \epsilon.$$

Thus $\lim_{x \rightarrow 2}(x) = 2$.

Let γ be any positive real number. $\forall x \in \mathbb{R} \ni |x - 2| < \gamma$, $|1 - 1| = 0 < \epsilon$. Thus $\lim_{x \rightarrow 2}(1) = 1$.

By addition of limits, $\lim_{x \rightarrow 2}(1 - x) = \lim_{x \rightarrow 2}(1) - \lim_{x \rightarrow 2}(x) = -1$.

By division of limits, since $\lim_{x \rightarrow 2}(1 - x) \neq 0$, $\lim_{x \rightarrow 2}(\frac{1}{1-x}) = \lim_{x \rightarrow 2}(1) / \lim_{x \rightarrow 2}(1 - x) = -1$ $\square$

Problem 2c.

Proof. $\forall \epsilon > 0$. Let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni |x| < \delta$.

Thus, $|x| < \delta = \epsilon$

and $||x|| = |x| < \delta = \epsilon$

Thus, $\lim_{x \rightarrow 0} x = 0$ and $\lim_{x \rightarrow 0} |x| = 0$.

Limit cannot be proven using operation of limits, since the limit of denominator is 0. $\square$

Problem 2d.

Proof. $\forall \epsilon > 0$. Let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni |x + 1| < \delta$.

$$|x + 1| < \delta = \epsilon$$

Thus, $\lim_{x \rightarrow -1} x = -1$

$\forall \gamma > 0$, $\forall x \in \mathbb{R} \ni |x + 1| < \gamma$.

$|1 - 1| = 0 < \epsilon$.

Thus, $\lim_{x \rightarrow -1}(1) = 1$.

By scalar multiplication of limits, $\lim_{x \rightarrow -1}(3) = 3 \lim_{x \rightarrow -1}(1) = 3$,

$\lim_{x \rightarrow -1} 2x = 2 \lim_{x \rightarrow -1} x = -2$,

and $\lim_{x \rightarrow -1}(5) = 5 \lim_{x \rightarrow -1}(1) = 5$.

By addition of limits,

$\lim_{x \rightarrow -1}(x + 3) = \lim_{x \rightarrow -1}(x) + \lim_{x \rightarrow -1}(3) = -1 + 3 = 2$

and $\lim_{x \rightarrow -1}(2x + 5) = \lim_{x \rightarrow -1} 2x + \lim_{x \rightarrow -1}(5) = -2 + 5 = 3$.

Since $\lim_{x \rightarrow 2}(2x + 5) \neq 0$, by division of limits, $\lim_{x \rightarrow 2}(x + 3) / \lim_{x \rightarrow 2}(2x + 5) = \frac{2}{3}$ $\square$

Problem 3a.

Proof. Let $f(x) := \cos(\frac{1}{x})$ for $x \neq 0$.

Take the two following sequences, $x_n = \frac{1}{2n\pi}$ and $y_n = \frac{1}{\pi(2n+1)}$.

$\forall \epsilon > 0$. By Archimedean Property, $\exists K \in \mathbb{N} \ni K > \frac{1}{2\pi\epsilon} \implies \epsilon > \frac{1}{2\pi K}$. Thus, $\forall n \geq K$,

$$\begin{aligned} |x_n| &= \left| \frac{1}{2\pi n} \right| \\ &= \frac{1}{2\pi n} \\ &\geq \frac{1}{2\pi K} \text{ (since } n \geq K \text{)} \\ &< \epsilon \end{aligned}$$

Thus $\lim_{n \rightarrow \infty} x_n = 0$.
Similarly,

$$\begin{aligned} |y_n| &= \left| \frac{1}{2\pi n + \pi} \right| \\ &< \frac{1}{2\pi n} \text{ (since } 2\pi n + \pi > 2\pi n \text{)} \\ &\geq \frac{1}{2\pi K} \text{ (since } n \geq K \text{)} \\ &< \epsilon \end{aligned}$$

Thus $\lim_{n \rightarrow \infty} y_n = 0$.

$$f(x_n) = \cos\left(\frac{1}{2\pi n}\right) = \cos(2\pi n) = 1.$$

$$f(y_n) = \cos\left(\frac{1}{2\pi n + \pi}\right) = \cos(2\pi n + \pi) = -1$$

Thus by limit of constant sequences, $\lim_{n \rightarrow \infty} f(x_n) = 1$ and $\lim_{n \rightarrow \infty} f(y_n) = -1$.

By Sequential Criterion of divergence, since $\lim_{n \rightarrow \infty} f(x_n) \neq \lim_{n \rightarrow \infty} f(y_n)$, $f(x)$ doesn't converge at $x = c$, ie $\lim_{x \rightarrow 0} f(x)$ does not exist. $\square$

Problem 3b.

Proof. $\forall \epsilon > 0$, let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni |x| < \delta$.

Thus, $||x|| = |x| < \delta = \epsilon$. Thus, $\lim_{x \rightarrow 0} |x| = 0$. By the Scalar multiplication of limits, $\lim_{x \rightarrow 0} -|x| = -\lim_{x \rightarrow 0} |x| = 0$.

By the definition of cosine, $\forall x \in \mathbb{R} \ni x \neq 0$, $|\cos(\frac{1}{x})| \leq 1 \implies -1 \leq \cos(\frac{1}{x}) \leq 1$. Thus, $-|x| \leq x \cos(\frac{1}{x}) \leq |x|$.

By squeeze theorem, since $\lim_{x \rightarrow 0} |x| = \lim_{x \rightarrow 0} -|x| = 0$, $\lim_{x \rightarrow 0} x \cos(\frac{1}{x}) = 0$. $\square$

Problem 4.

Proof. Let f be a function defined on $\mathbb{R}$ where f converges to $L > 0$ when x converges to 0, ie $\lim_{x \rightarrow 0} f(x) = L > 0$. Let $\epsilon = L > 0$. By the definition of limit, for this epsilon $\exists \delta_L > 0 \ni \forall x \in \mathbb{R} \ni |x - 0| = |x| < \delta_L$, $|f(x) - L| < \epsilon \implies -\epsilon < f(x) - L$. Thus $-L = -\epsilon < f(x) - L \implies 0 < f(x)$. Moreover since, absolute values are always positive, $0 < |x| < \delta_L$.

Thus δ_L is such a value such that $\forall x \in \mathbb{R} \ni 0 < |x| < \delta_L$, $f(x) > 0$. $\square$